

Lecture #17,18

Content to be covered

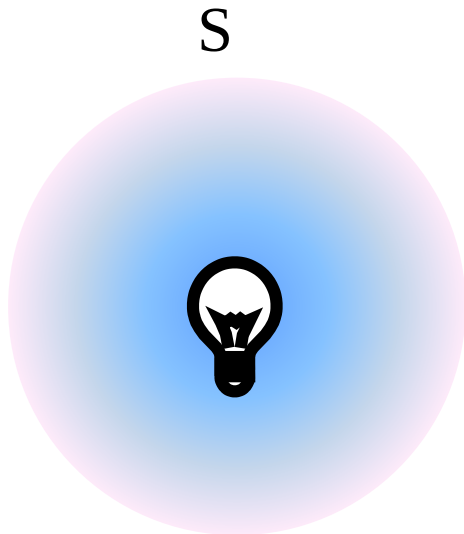
- ❖ Constancy of the Speed of Light
- ❖ Lorentz Transformation
- ❖ Consequence of Relativity
 - ❑ Length Contraction
 - ❑ Time Dilation
 - ✓ Muon decay problem
 - ✓ Twin Paradox

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

The Constancy of the Speed of Light

- Consider the fixed system S and the moving system S' .
- At $t = 0$, the origins and axes of both systems are coincident with system S' moving to the right along the x axis.
- A flashbulb goes off at both origins when $t = 0$.
- According to postulate 2, the speed of light will be c in both systems and the wave fronts observed in both systems must be spherical.



Spherical wavefronts in S:

$$x^2 + y^2 + z^2 = c^2 t^2$$

Spherical wavefronts in S':

$$x'^2 + y'^2 + z'^2 = c^2 t'^2$$

Note that this cannot occur in Galilean transformations.

$$x' = x - vt$$

$$y' = y$$

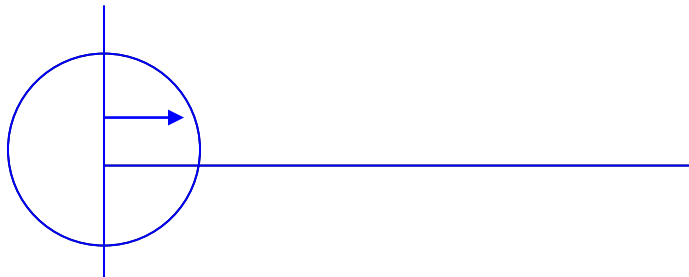
$$z' = z$$

$$t' = t$$

$$x'^2 + y'^2 + z'^2 = (x^2 - 2xvt + v^2 t^2) + y^2 + z^2 \neq c^2 t'^2$$

There are a couple of extra terms
(~~$-2xvt + v^2 t^2$~~) in the primed frame.

S



S'

The Lorentz Transformation

The special set of linear transformations that:

preserve the constancy of the speed of light (c) between inertial observers; with the assumption that the time is not an absolute quantity.

Therefore $t \neq t'$

known as the Lorentz transformation equations

The Lorentz Transformation: Derivation

Consider a system of two inertial frames of references S and S' .

S : is at rest and S' : is a moving frame with constant velocity $\vec{v}$ relative to S

$(x, y, z, t) \rightarrow$ coordinates of event P for an observer O in the stationary frame S

$(x', y', z', t') \rightarrow$ coordinates of event P for an observer O' in the moving frame S'

Let $x' = \gamma(x - vt)$ (1) γ : proportionality constant.

According to first postulate of STR

Therefore $x = \gamma(x' + vt')$ (2) where $t \neq t'$

Here γ is a factor of proportionality which does not depend upon either x or t but may be a function of v . γ must be same in both the reference frames as there is no difference between S and S' other than the sign of v .

According to second postulate of STR

The wavefront along x, x' axis must satisfy

$x = ct$ and $x' = ct'$

Thus substitute the value of x and x' in eq (1) and (2) respectively.

$ct' = \gamma(c - v)t$ and $ct = \gamma(c + v)t'$

Multiply these equations and solve it for γ

$$\gamma = \frac{1}{\sqrt{1 - v^2 / c^2}}$$

The Lorentz Transformation: Derivation

$$x' = \gamma(x - vt) \quad \dots(1)$$

$$x = \gamma(x' + vt') \quad \dots(2)$$

From eq(1) and eq(2)

$$x = \gamma[\gamma(x - vt) + vt'] \quad \text{or} \quad \frac{x}{\gamma} = \gamma(x - vt) + vt'$$

$$\therefore t' = \frac{x}{\gamma v} - \frac{\gamma x}{v} + \gamma t \quad \text{or} \quad t' = \gamma t - \frac{\gamma x}{v} \left(1 - \frac{1}{\gamma^2}\right)$$

similarly $t = \gamma t' - \frac{\gamma x'}{v} \left(1 - \frac{1}{\gamma^2}\right)$

$$t' = \gamma \left[t - \frac{vx}{c^2} \right] = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$t = \gamma \left[t' + \frac{x'v}{c^2} \right] = \frac{t' + \frac{x'v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Imp: Prove that

$$x^2 + y^2 + z^2 - c^2 t^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$$

under Lorentz Transformation

Lorentz Transformation

Direct

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Inverse

$$x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$y = y'$$

$$z = z'$$

$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Handwritten notes:
 $\downarrow$ $dx' = \gamma(dx - v dt)$
 $\downarrow$ $dx = \gamma(dx' + v dt')$

- Measurements of time and position depend upon the frame of reference of observer so that two events which occur simultaneously in one frame at different places need not be simultaneous in another.

- Lorentz transformation reduces to ordinary Galilean transformation when $v \ll c$.

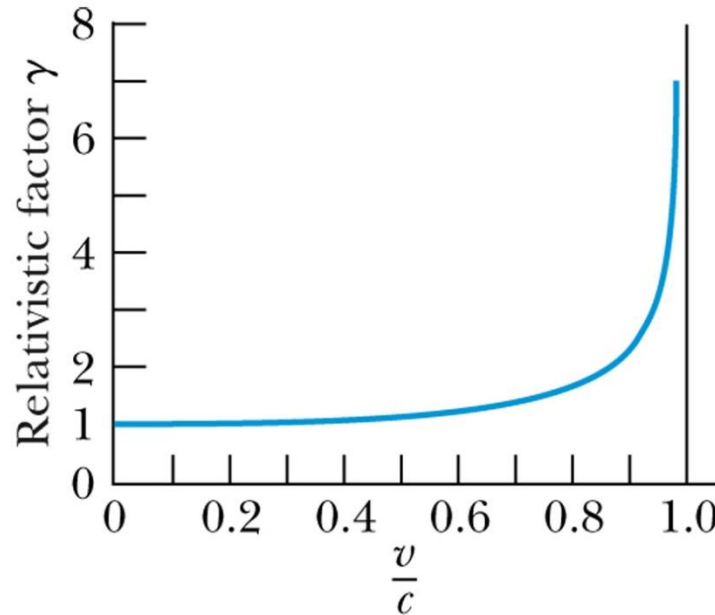
The complete Lorentz Transformation

$$x' = \frac{x - vt}{\sqrt{1 - v^2 / c^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - vx / c^2}{\sqrt{1 - v^2 / c^2}}$$



Approximate Values for γ

$$x = \frac{x' + vt'}{\sqrt{1 - v^2 / c^2}}$$

$$y = y'$$

$$z = z'$$

$$t = \frac{t' + vx' / c^2}{\sqrt{1 - v^2 / c^2}}$$

0.995
0.999

10.01
22.37

If $v \ll c$, i.e. $\gamma \approx 1$, yielding the familiar Galilean transformation.

Space and time are now linked, and the frame velocity cannot exceed c .

Consequence of Relativity

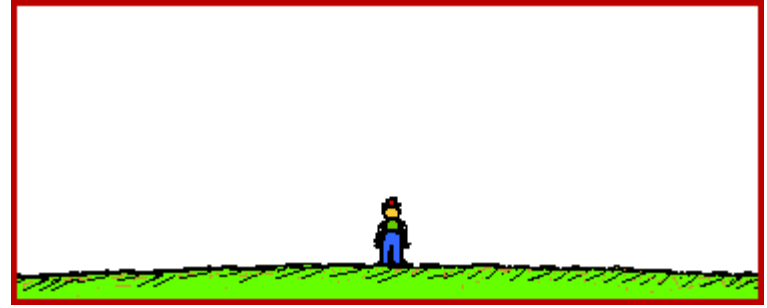
➡ The Major Consequences To This Theory are:-

- Ⓢ Length Contraction
- Ⓢ Time Dilation
- Ⓢ Mass Expansion

Lorentz Contraction

A fast-moving plane at different speeds.

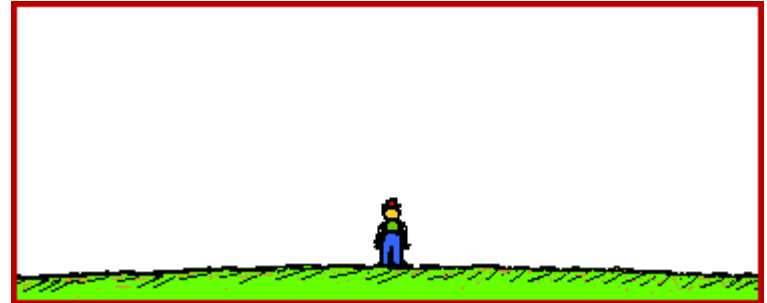
$$v = 10\% c$$



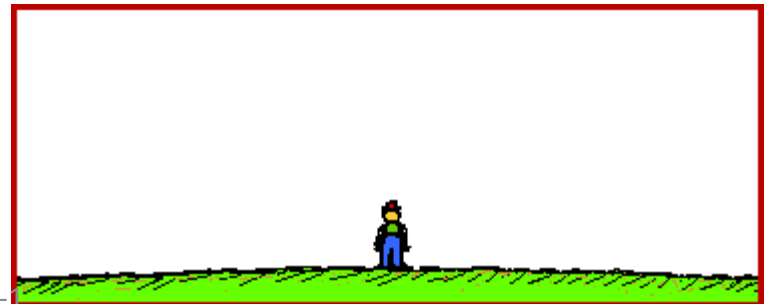
$$v = 80\% c$$



$$v = 99\% c$$



$$v = 99.9\% c$$



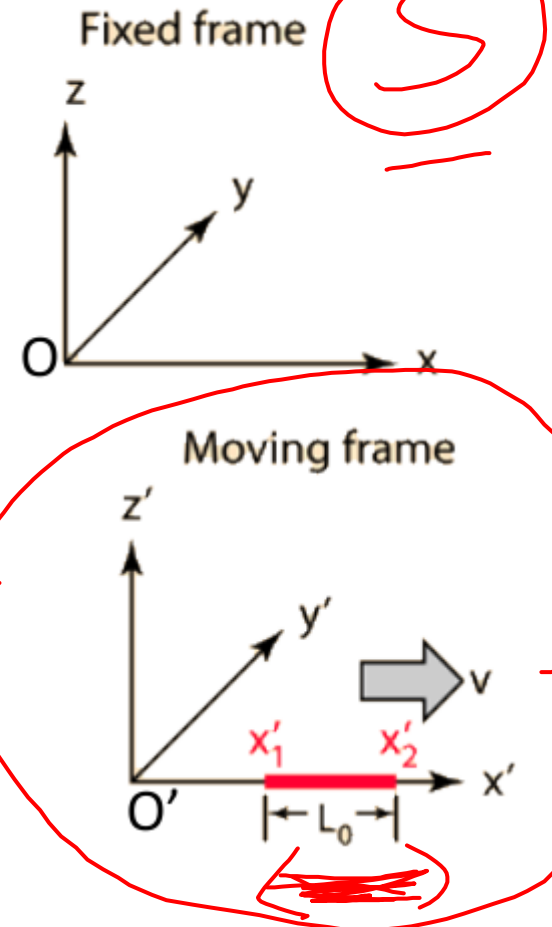
Length Contraction

A rod is lying along the x' axis of the moving frame S' . If an observer determines the coordinates of its ends to be x'_1 and x'_2 , then the length of the rod w.r.t. O' is: $L_0 = x'_2 - x'_1$

Therefore, L_0 is the actual length of the rod (w.r.t. to O') in a frame in which the rod is at rest. According to Lorentz transformation, length measured in S w.r.t. O will be the apparent length ($L = x_2 - x_1$).

$$x'_1 = \gamma(x_1 - vt) \quad x'_2 = \gamma(x_2 - vt)$$

$$L_0 = \gamma(x_2 - vt) - \gamma(x_1 - vt) = \gamma(x_2 - x_1) = \frac{L}{\sqrt{1 - \frac{v^2}{c^2}}}$$



As ends of the rod is measured at the same time, so; $t_1 = t_2$ in S .

Length Contraction

Therefore $L = L_0 \sqrt{1 - v^2 / c^2} = \frac{L_0}{\gamma}$

The length (L_0) of an object in motion with respect to an observer (in S Frame) appears to be shorter, this phenomenon is known as **Lorentz FitzGerald contraction** or **Length Contraction**.

Faster means shorter

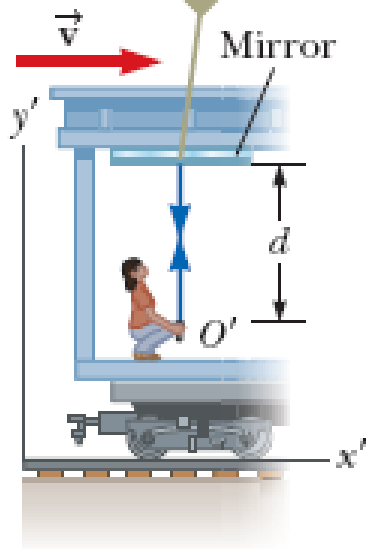
Case I : if $v=0$, $L = ?$

Case II: if $v=c$, $L = ?$

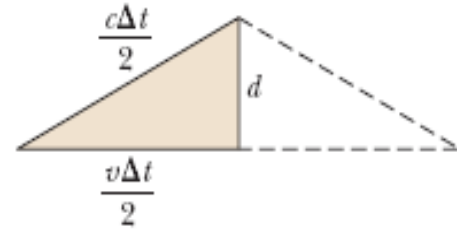
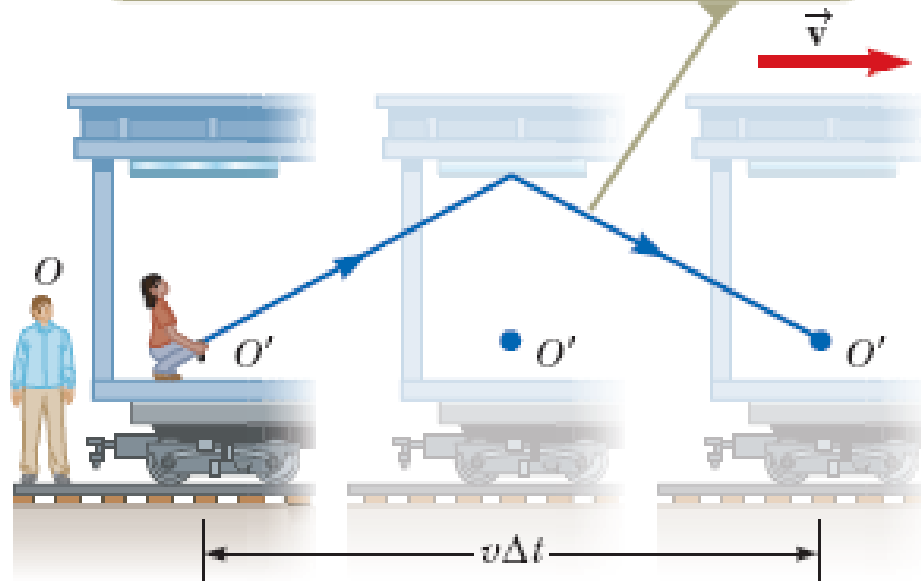
Exercise : Find L when $L_0=1\text{m}$ and $v=0.9c$.

Time Dilation

Observer O' sees the light pulse move up and down vertically a total distance of $2d$.



Observer O sees the light pulse move on a diagonal path and measures a distance of travel greater than $2d$.



$$\left(\frac{c\Delta t}{2}\right)^2 = \left(\frac{v\Delta t}{2}\right)^2 + d^2$$

$$t = \frac{\Delta t_p}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma \Delta t_p$$

$$\Delta t_p = \frac{\text{distance traveled}}{\text{speed}} = \frac{2d}{c}$$

$\therefore \gamma > 1, \text{ thus } \Delta t > \Delta t_p$

Time Dilation

Assume a clock is placed at x' in the moving frame S' . When an observer in S' measures a time interval

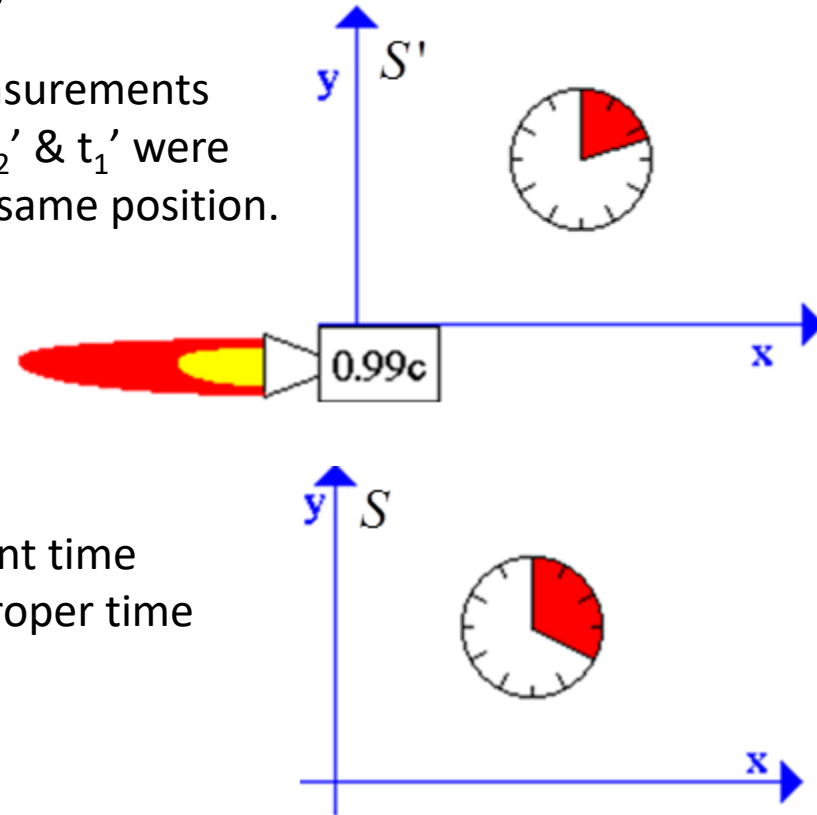
$$t_o = t'_2 - t'_1 \quad \text{proper time}$$

Two measurements of time t'_2 & t'_1 were taken at same position.

The observer in S , will measure this interval as

$$\begin{aligned} t = t_2 - t_1 &= \frac{t'_2 + \frac{v x'_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t'_1 + \frac{v x'_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{t_o}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

apparent time
or improper time



Thus improper time will be longer than proper time; called **Time dilation**.

Time Dilation

$$t = \gamma t_0$$

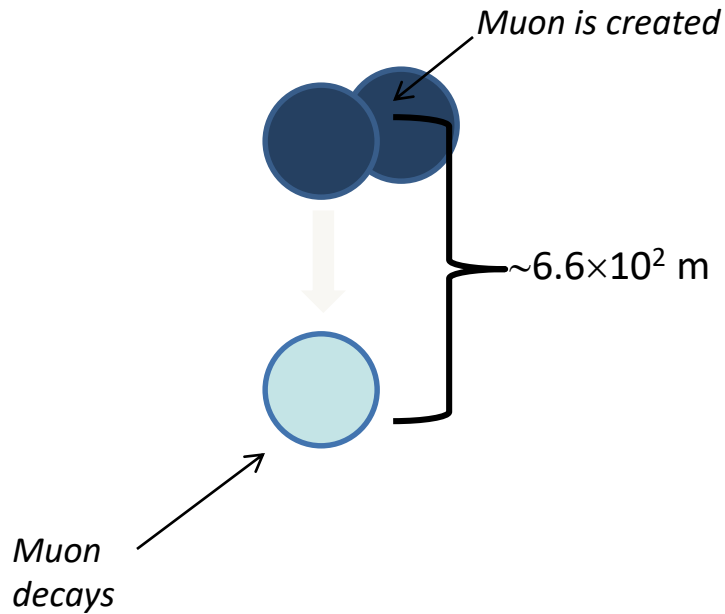
A moving clock ticks more slowly than a clock at rest.

Thus a stationary clock measures a longer time interval between events occurring in a moving frame of reference – time dilation.

Since $\gamma \geq 1$, t will be larger than t_0 .

Thus if the time interval between two beats (tick) of the clock in the proper frame S' is 1 second, then this time interval will be observed more than 1 second in the frame S *i.e* a **moving clock appears to go slow.**

Example of Time Dilation



The life time of μ mesons is $2.2 \mu\text{s}$ and their speed $0.998c$, so that they can cover only a distance of $0.998c \times 2.2 \mu\text{s}$ or 0.66km in their life time, and yet they are found in abundance at sea level, *i.e.*, at a depth of 10 km from the upper atmosphere where they are produced. How may this be explained on the basis of (i) length contraction (ii) time dilation?

Example of Time Dilation

Muon is created



(i) Length contraction:

$$t_p = 2.2\mu s, L = 2.2\mu s \times 0.998c = 0.66km$$

Distance on the earth,

$$L = L_o \sqrt{1 - v^2 / c^2} \Rightarrow \boxed{L_o = 10.42km > 10km}$$

(ii) Time Dilation

$$t = \frac{t_p}{\sqrt{1 - v^2 / c^2}} \Rightarrow t = 34.8\mu s$$

$\therefore$ distance travelled on the earth

$$= 34.8\mu s \times 0.998c = \boxed{10.42km > 10km}$$

Muon decays



Problem-2

1. An observer in the laboratory sees two particles colliding at $x = 20.5 \text{ m}$, $y=0$, $z=0$ and $t= 7.2 \text{ s}$. What are the co ordinates of this event in a frame moving at 30.5 m/s with respect to the laboratory frame?
2. A particle with a proper life time of $1 \text{ }\mu\text{s}$ moves through the laboratory at $0.9c$. What is its lifetime as measured by observers in the laboratory?
Ans.: $2.294 \text{ }\mu\text{s}$.

Twin Paradox

A longer life but it will not seem longer!

This paradox involves two identical clocks; one remains on the earth while other one is taken into space craft at speed v and brought back to earth.

Assume a pair of twin like Dick and Jane both are of 20yrs old. Dick takes off on a voyage with speed $0.8c$ to a star 20 light years away and then back to earth. Jane stayed behind on the earth. Dick returns after 50 yrs according to Jane's calendar. Means, Jane is 70 yrs old. And Dick, who moved to space with $0.8 c$, is 50 yrs old only according to:

$$t = \frac{t_o}{\sqrt{1 - v^2/c^2}} = t_o / 0.60 \Rightarrow$$

$t_o / \sqrt{1 - v^2/c^2} = t \Rightarrow t_o = 0.60 \times 50 \text{ yrs} = 30 \text{ yrs}$, i.e. Dick's biological decay was only 60%, **it means, Dick appears to be younger as compared to his twin brother left on the earth.**